Supplementary Material S2

Independent validation of the AI-disclosure classifier

Companion to *Disclosed Intelligence: A Large-Sample Measurement of AI Disclosure in U.S. Investment Adviser Fiduciary Filings*. This supplement documents the validation sample, the coding protocol, per-label agreement and error metrics, the full disagreement analysis with reasoning, and the implication of the measured error for the headline prevalence estimate. All scoring code and the blinded coding sheet are deposited with the replication package.

S2.1 Design and rationale

The main text distinguishes three levels of evidence about the classifier. Inter-model reproducibility (Table 4, same-family) asks whether a second model from the same family reproduces the primary labels; it establishes stability but not accuracy, because two models trained on overlapping data can share the same systematic error. Independent validation — reported here — asks whether a rater that did not produce the primary labels, and that comes from a different model family, reaches the same judgments when working blind from the brochure text and the frozen rubric alone. This is the stronger reliability test: it runs over a larger sample (180 vs. 60 brochures), it is fully blinded to the primary labels, and it crosses model families, so shared-family artifacts cannot inflate agreement.

We are explicit about what the independent rater is. The primary classifier is OpenAI gpt-4o. The independent rater is a large language model from a **different developer and family** (Anthropic Claude), operated as a careful rubric-based coder: each brochure was read from keyword-windowed excerpts, each of the five labels was assigned only when supported by a verbatim quote about the filing firm's own practices, and the rater had no access to the primary labels or any summary of them. Because both instruments are language models, the exercise measures cross-family, cross-instrument agreement against a common written rubric, which bounds instrument-induced error; it is not a substitute for a panel of human domain experts. We therefore report the residual disagreement set in full (S2.4) so that a human audit — the final recommended step — can be conducted against a small, enumerated list rather than the whole corpus.

S2.2 Validation sample

The validation set is the 180 brochures defined in the coding sheet: **all 123 brochures the primary classifier flagged with at least one label**, plus a **random sample of 57 brochures the classifier scored all-negative** (seed 42), drawn to estimate the classifier's miss rate as well as its false-positive rate. Enriching on model-positives is deliberate — a uniform random sample of 180 would contain too few positives to estimate precision on the rare labels — but it means the raw positive rate within the validation set is not a population prevalence estimate; the population estimate comes from the full 388-brochure classification in the main text. The validation set spans both adviser types and all four AUM quartiles.

S2.3 Agreement and error metrics

For each label we treat the independent blind re-coding as the reference. Table S2.1 gives the confusion counts and Table S2.2 the derived error metrics. PABAK is reported alongside κ because several labels have low prevalence, where κ is unstable. The derived any-use indicator (a ∪ b ∪ e) — the estimand behind the headline number and the survey wedge — is scored as its own row. For every cell n = 180; “Model +” and “Ind. +” are the positives assigned by each rater; FP = classifier-positive/independent-negative and FN = classifier-negative/independent-positive.

Label / indicator	Model +	Ind. +	TP	FP	FN	TN
(a) investment process	71	63	49	22	14	95
(b) operations / client service	26	31	21	5	10	144
(c) disclosed risk factor	107	99	98	9	1	72
(d) explicit non-use	8	15	7	1	8	164
(e) named vendor	21	11	11	10	0	159
Any use (a ∪ b ∪ e)	92	80	70	22	10	78

Table S2.1. Confusion counts for the primary classifier (gpt-4o) against the independent blind re-coding; n = 180.

Label / indicator	Prec.	Rec.	F1	% agree	κ	PABAK
(a) investment process	0.690	0.778	0.731	80.0%	0.573	0.600
(b) operations / client service	0.808	0.677	0.737	91.7%	0.688	0.833
(c) disclosed risk factor	0.916	0.990	0.951	94.4%	0.887	0.889
(d) explicit non-use	0.875	0.467	0.609	95.0%	0.585	0.900
(e) named vendor	0.524	1.000	0.688	94.4%	0.660	0.889
Any use (a ∪ b ∪ e)	0.761	0.875	0.814	82.2%	0.645	0.644

Table S2.2. Precision, recall, F1, percent agreement, Cohen's κ and PABAK for the primary classifier with the independent blind re-coding as reference; n = 180.

Three patterns follow. First, the constructs the main text treats as reliable are confirmed on this larger, cross-family, blinded sample: the risk-factor label reaches κ = 0.887 (F1 = 0.951) and the named-vendor label recovers every independent positive (recall = 1.000). Second, the any-use estimand shows substantial agreement (κ = 0.645) with recall 0.875 and precision 0.761 — the classifier finds seven of every eight independently-confirmed adopters, at the cost of roughly one false positive in four flags. Third, the finer use labels are the softest: investment-process (a) is the lowest-κ label (0.573), consistent with the inter-model result and with the substantive boundary problem documented next.

S2.4 Disagreement analysis

Sixty of 180 brochures (33%) differ on at least one label; 32 differ on the derived any-use indicator (22 classifier-positive/independent-negative, 10 classifier-negative/independent-positive). The disagreements are not noise — they cluster on four interpretable boundary cases, and the two directions partly offset.

Classifier over-calls investment-process AI (a) in four recurring situations. (1) Reserve-the-right-to-use language: several brochures disclose only that the firm may or reserves the right to use AI in future, with no present use; the classifier read the capability as use, the independent coder scored these as risk-factor-only (examples: The Sterling Group, Glade Brook Capital Partners, Ivy Hill Asset Management, Shorehill Capital, Bruckmann Rosser Sherrill, Cinven). (2) Third-party or execution algorithms: broker smart-order-routing, Section 28(e) execution software, and an IntraFi FDIC-sweep “proprietary algorithm” are tools used via vendors, not the firm's own AI investing (Ameriprise, DWS Investments Australia, Bayard Asset Management). (3) Rule-based robo-advice: rule-based model-portfolio-matching software is plain automation not framed as AI/ML in the text (Acorns, John Hancock Personal Financial Services, a Betterment-based program). (4) Risk-factor-only firms: where AI appears only as a market, competitive, or portfolio-company risk, the classifier occasionally added (a) on top of the correct (c) (Recurring Capital, Shorehill, McGinty Road Partners).

Classifier over-calls named-vendor (e) when a product is named illustratively. In several brochures "OpenAI ChatGPT" is named only as a generic example of a market-wide technology, not a tool the firm uses (Bain Capital Public Equity, McGinty Road Partners, WPH GP). The classifier scored (e); the independent coder required the named product to be invoked as the firm's own capability. This is why named-vendor has high recall but modest precision (0.524), and why the main text reports it separately from any-use.

Classifier under-calls investment-process AI (a) on genuine quantitative-model firms. In the opposite direction, ten brochures describe the firm's own quantitative or algorithmic model-driven investment process — which the rubric's scope clause explicitly counts as in-scope — that the classifier missed (Voyager Capital Management, Ionic Capital Management, Curtis Advisory Group, Vis Advisors, Roubaix Capital, Wisdom Capital, S Squared Technology, Breed's Hill Capital, Cadent Capital Advisors). Here the independent coder was the more inclusive rater. This AI/quantitative-model boundary is the single largest source of the (a)-label disagreement.

Explicit non-use (d) is rare and the classifier missed roughly half of the independently-identified non-use statements (recall 0.467); because the prevalence is tiny, PABAK is high (0.900) but κ moderate (0.585). Non-use disclosures are often a single sentence embedded in a code-of-ethics or compliance passage that keyword windowing can truncate.

S2.5 Implication for the headline estimate

The two error directions on any-use partly cancel. Within the validation set the classifier flags 92 any-use brochures against the independent coder's 80 (ratio 0.87). Applying that ratio to the design estimate as a first-order bias correction moves disclosed any-use from **23.7% to roughly 20.6%**, which remains inside the reported design-based 95% confidence interval (19.7–28.2). The correction is small because the false positives (reserve-the-right language, illustrative ChatGPT mentions, third-party algorithms) are substantially offset by the false negatives (missed own-firm quantitative-model investing). Two conclusions are unchanged and, if anything, reinforced. The level of disclosed AI use is low — a modest downward correction leaves it near one in five brochures. And the survey-versus-disclosure wedge is not an artifact of classifier over-counting: correcting the classifier's known over-calls lowers disclosed use and therefore widens the gap against the ~63% surveyed-adoption benchmark. The most conservative reading is that the wedge is real and the fine investment-process label should be read as approximate, with named-vendor and any-use as the load-bearing constructs.

S2.6 Limitations of the validation

The independent rater is a language model, not a panel of human domain experts; the exercise establishes cross-family, cross-instrument agreement against a written rubric, which bounds instrument-specific error but cannot by itself certify human-level accuracy. Both raters read the same keyword-windowed excerpts, so error introduced by windowing (e.g., truncated non-use statements) is shared rather than measured. The validation set is enriched on classifier-positives, so precision on the rare labels is estimated more sharply than recall on the full population. The residual 60-brochure disagreement set is enumerated in the replication file (P10A_S2_disagreements.csv) so that a targeted human audit can adjudicate the boundary cases — reserve-the-right language, third-party vs. own-firm algorithms, and the quantitative-model/AI line — that drive essentially all of the classifier's measured error.

References

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